Nearby cycles over general bases

郑维喆

Artin stack of f.p.

$$\begin{tikzcd}
RSht|_{\Delta(\bar{x})} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & RSht \times_{X^I} X \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & RSht \arrow[d] \\
\bar{x} \arrow[r] & X \arrow[r, "\Delta_d"] & X^I \\
\bar{\eta} \arrow[u] & &
\end{tikzcd}$$

$L \in D^b_c(RSht, \Lambda)$

$\mathcal{F}_d = \Psi(\Delta_d^* L) \in D^b_c(RSht|_{\Delta(\bar{x})}, \Lambda)$

$T_\delta : \mathcal{F}_d \longrightarrow \mathcal{F}_{d'}$, $\delta \in FWeil(k^{1,\dots,k}, \overline{k^{1,\dots,k}})$ $\quad k = \# I$

of deg $d' - d$

$d, d' \in \mathbb{Z}^I$

$\mathcal{F}_d = \Psi^{naive}_{\Delta(\bar{x}) \leftsquigarrow \Delta_d(\bar{\eta})}(L)$

$$\begin{tikzcd}
\overline{\eta_{I,d}} \arrow[r, phantom, "\overset{\delta}{\simeq}"] \arrow[d, squiggly] & \overline{\eta_{I,d'}} \arrow[d, squiggly] \\
\Delta_d(\bar{\eta}) & \Delta_{d'}(\bar{\eta}) \\
\Delta(\bar{x}) \arrow[u, squiggly] \arrow[ur, squiggly] &
\end{tikzcd}$$

$\delta \in FWeil(\eta_I, \bar{\eta}_I)$

$\overline{\eta_{I,d}} = \prod Frob_{\{i\}}^{d_i}(\bar{\eta}_I)$

$$\begin{tikzcd}
\Psi_{\Delta(\bar{x}) \leftsquigarrow \Delta_d(\bar{\eta})} \arrow[r, "sp^*"] \arrow[d, "res"] & \Psi_{\Delta(\bar{x}) \leftsquigarrow \overline{\eta_{I,d}}} \arrow[r, phantom, "\overset{\delta^*}{\simeq}"] & \Psi_{\Delta(\bar{x}) \leftsquigarrow \overline{\eta_{I,d'}}} & \Psi_{\Delta(\bar{x}) \leftsquigarrow \Delta_{d'}(\bar{\eta})} \arrow[l, "sp^*"'] \arrow[d, "res"] \\
\Psi^{naive}_{\Delta(\bar{x}) \leftsquigarrow \Delta_d(\bar{\eta})} & & & \Psi^{naive}_{\Delta(\bar{x}) \leftsquigarrow \Delta_{d'}(\bar{\eta})} \\
\mathcal{F}_d \arrow[u, phantom, "\shortparallel"] \arrow[rrr, dashed, "T_\delta"] & & & \mathcal{F}_{d'} \arrow[u, phantom, "\shortparallel"]
\end{tikzcd}$$

isom. in good cases

$$\begin{tikzcd} L \in D^+(X, \Lambda) \arrow[d, dash] & \\ X \arrow[d] & \\ S & \end{tikzcd}$$

$X \to S$ morphism of schemes, Λ fin. comm. ring, $n\Lambda = 0$, n invertible on S.

$\bar{s} \leftsquigarrow \bar{t}$ on S

$$\begin{tikzcd} X_{\bar{t}} \arrow[r, hook, "j^{\text{naive}}"] \arrow[d] \arrow[dr] & X_{(\bar{s})} \arrow[d] & X_{\bar{s}} \arrow[l, hook', "i"'] \arrow[d] \\ & X_{(\bar{t})} \arrow[u, "j"] \arrow[d] & \\ \bar{t} \arrow[r] \arrow[dr] & S_{(\bar{s})} & \bar{s} \arrow[l, hook'] \\ & S_{(\bar{t})} \arrow[u] & \end{tikzcd}$$

$$\Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{t}}(L) := i^* j^{\text{naive}}_* (L|X_{\bar{t}}) \in D^+(X_{\bar{s}}, \Lambda)$$

$$\uparrow \text{res}$$

$$\Psi_{\bar{s} \leftsquigarrow \bar{t}}(L) := i^* j_* (L|X_{(\bar{t})}) \in$$

Rem. - t generic point, $\Psi_{\bar{s} \leftsquigarrow \bar{t}} = \Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{t}}$

- In general, $T =$ closure of $\bar{t}$ in $S_{(\bar{s})}$

$$\Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{t}}(L) = \Psi_{\bar{s} \leftsquigarrow \bar{t}}(L|X_T)$$

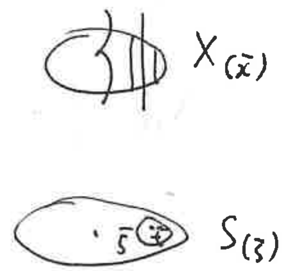
Functoriality. $\bar{s} \leftsquigarrow \bar{t} \leftsquigarrow \bar{u}$

$$\begin{tikzcd} S_{(\bar{u})} \arrow[r] & S_{(\bar{t})} \arrow[r] & S_{(\bar{s})} \\ \bar{u} \arrow[u] & \bar{t} \arrow[u] & \end{tikzcd}$$

$$\begin{tikzcd} \Psi_{\bar{s} \leftsquigarrow \bar{t}} \arrow[r, "sp^*"] \arrow[d, "\text{res}"'] & \Psi_{\bar{s} \leftsquigarrow \bar{u}} \arrow[d, "\text{res}"] \\ \Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{t}} & \Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{u}} \end{tikzcd}$$

$$\left(\Psi^{\text{naive}}_{\bar{s} \leftarrow \bar{t}} L\right)_{\bar{x}} \simeq R\Gamma\left(X_{(\bar{x})} \times_{S_{(\bar{s})}} \bar{t}, L\right)$$

$$\left(\Psi_{\bar{s} \leftarrow \bar{t}} L\right)_{\bar{x}} \simeq R\Gamma\left(X_{(\bar{x})} \times_{S_{(\bar{s})}} S_{(\bar{t})}, L\right)$$



X/S sep. of. f.t.

$$\begin{array}{ccc}
R\Gamma_c\left(\Psi_{\bar{s} \leftarrow \bar{t}}\right) & \xrightarrow{sp^*} & R\Gamma_c\left(\Psi_{\bar{s} \leftarrow \bar{u}}\right) \\
\downarrow \text{res} & & \downarrow \text{res} \\
R\Gamma_c\left(\Psi^{\text{naive}}_{\bar{s} \leftarrow \bar{t}}\right) & & R\Gamma_c\left(\Psi^{\text{naive}}_{\bar{s} \leftarrow \bar{u}}\right) \\
\downarrow \text{BC} & & \downarrow \text{BC} \\
R\Gamma_c(X_{\bar{t}}, L) & \xrightarrow{sp^*} & R\Gamma_c(X_{\bar{u}}, L)
\end{array}$$

Ex. $k = \bar{k}$, $\quad \begin{array}{c} X = Bl_0(S) \\ \downarrow \\ S = (\mathbb{A}^2_k)_{(0)} \end{array}$ $\quad \bar{x} \in X_0$

$0 \leftarrow \bar{t}$, $\quad t \neq 0$

$$X_{(\bar{x})} \times_S S_{(\bar{t})} \simeq X_{(\bar{x})} \times_X X_{(\bar{t})}$$

$$H^i\left(X_{(\bar{x})} \times_X X_{(\bar{t})}, \Lambda\right) = 0, \quad \forall\, i > 0.$$

$$\# \pi_0\left(X_{(\bar{x})} \times_X X_{(\bar{t})}\right) = \infty, \qquad \dim H^0\left(X_{(\bar{x})} \times_X X_{(\bar{t})}, \Lambda\right) = \infty.$$

$\Psi_{0 \leftsquigarrow \bar{t}} \Lambda$ not constructible

<u>Case</u> $t = \eta$, $\Psi_{0 \leftsquigarrow \bar{t}} \Lambda = \Psi^{\text{naive}}_{0 \leftsquigarrow \bar{t}} \Lambda$ not constructible.

<u>Case</u> $t \neq \eta$. $\underbrace{\Psi_{0 \leftsquigarrow \bar{t}} \Lambda \xrightarrow{\text{res}}}_{\text{not isom.}} \underbrace{\Psi^{\text{naive}}_{0 \leftsquigarrow \bar{t}} \Lambda}_{\text{constructible}}$

If x not in closure of t in X, $X_{(\bar{x})} \times_S \bar{t} = \phi$.

<u>Th.</u> (Orgogozo) S qcqs. has finitely many irred. components (eg S noeth.)

f of finite presentation, $L \in D^b_c(X; \Lambda)$, Then $\exists$ <u>proper birational morphism</u> (modification)

$\tilde{S} \longrightarrow S$, such that

(1) $\forall\ \bar{s} \leftsquigarrow \bar{t}$ on $\tilde{S}$, $\tilde{X} = X \times_S \tilde{S}$

$$\begin{array}{c} \Psi_{\bar{s} \leftsquigarrow \bar{t}} L|_{\tilde{X}} \in D^b_c(\tilde{X}_{\bar{s}}, \Lambda) \\ \text{res} \downarrow \wr \\ \Psi^{\text{naive}}_{\bar{s} \leftsquigarrow \bar{t}} (L|\tilde{X}) \end{array}$$

(2) $\exists$ stratification $\tilde{S} = \cup S_\alpha$ by locally closed constructible subsets, s.t.

$\underbrace{\bar{s} \leftsquigarrow \bar{t} \leftsquigarrow \bar{u}}_{\text{on } S_\alpha}$, $\Psi_{\bar{s} \leftsquigarrow \bar{t}} L \xrightarrow[\sim]{sp^*} \Psi_{\bar{s} \leftsquigarrow \bar{u}} L$

<u>Rem.</u> – smooth base change $\begin{array}{c} Y \\ \downarrow \text{sm.} \\ X \;- L \\ \downarrow \\ S \end{array}$ $(\Psi_{\bar{s} \leftsquigarrow \bar{t}} L)|_{Y_{\bar{s}}} \xrightarrow{\sim} \Psi_{\bar{s} \leftsquigarrow \bar{t}} (L|Y)$

$\Rightarrow$ Th. holds for X Artin stack of f.p. $/S$.

$$\begin{tikzcd} \eta \arrow[r] & U \arrow[r, hook, "\circ"] \arrow[dr, "\circ"] & \tilde{S} \arrow[d] \\ \operatorname{Spec} R \arrow[u, "\text{gen pt}"] \arrow[rr] \arrow[urr, dashed, "\exists !"] & & S \end{tikzcd}$$

R valuation ring

<u>Deligne's vanishing topos</u>.

Upshot : $R\Psi L \in D^+(X \overleftarrow{\times}_S S , \Lambda)$

$\uparrow$

vanishing topos.

$R\Psi_{\bar{s} \leftarrow \bar{t}} L$ are slices of $R\Psi L$.

<u>Def</u> (oriented product)

$$\begin{tikzcd} & T' \arrow[d, "m"] \arrow[ddl, bend right, "p'"'] \arrow[ddr, bend left, "q'"] & \\ & T \arrow[dl, "p"'] \arrow[dr, "q"] & \\ X \arrow[dr, "f"'] & \overset{\tau}{\Leftarrow} & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

$(T, p, q, \tau : fp \Leftarrow gq)$ topos

f, g morphism of topoi

$\forall (T', p', q', \tau')$,

$\exists (m, \alpha, \beta)$ unique up to unique isom s.t. diag comm.

$$\begin{tikzcd} X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

morphism of schemes

$$X_{\bar{e}t} \overleftarrow{\times}_{S_{\bar{e}t}} Y_{\bar{e}t} =: X \overleftarrow{\times}_S Y$$

$$\begin{tikzcd} & X_{\bar{e}t} \overleftarrow{\times}_{S_{\bar{e}t}} Y_{\bar{e}t} \arrow[dl] \arrow[dr] & \\ X_{\bar{e}t} \arrow[dr] & & Y_{\bar{e}t} \arrow[dl] \\ & S_{\bar{e}t} & \end{tikzcd}$$

Pages

site for $X \overleftarrow{\times}_S Y$. Obj. :

$$
\begin{tikzcd}
U \arrow[r] \arrow[d, "\text{ét}"'] \arrow[dr, phantom, "\circlearrowleft"] & W \arrow[d, "\text{ét}"] \arrow[dr, phantom, "\circlearrowright"] & V \arrow[l] \arrow[d, "\text{ét}"] \\
X \arrow[r] & S & Y \arrow[l]
\end{tikzcd}
$$

morphisms

$$
\begin{tikzcd}
U \arrow[r] \arrow[d] & W \arrow[d] & V \arrow[l] \arrow[d] \\
U' \arrow[r] \arrow[d] & W' \arrow[d] & V' \arrow[l] \arrow[d] \\
X \arrow[r] & S & Y \arrow[l]
\end{tikzcd}
$$

<u>covers</u> (1) $U_i \to U$ cover.

$$
\begin{tikzcd}
U_i \arrow[r] \arrow[d] & W \arrow[d, equal] & V \arrow[l] \arrow[d, equal] \\
U \arrow[r] & W & V \arrow[l]
\end{tikzcd}
$$

(2) $V_i \to V$ cover

$$
\begin{tikzcd}
U \arrow[r] \arrow[d, equal] & W \arrow[d, equal] & V_i \arrow[l] \arrow[d] \\
U \arrow[r] & W & V \arrow[l]
\end{tikzcd}
$$

(3)

$$
\begin{tikzcd}
U \arrow[r] \arrow[d, equal] & W' \arrow[d] \arrow[dr, phantom, "\lrcorner"] & V' \arrow[l] \arrow[d] \\
U \arrow[r] & W & V \arrow[l]
\end{tikzcd}
$$

$Y = S$

$$
\begin{tikzcd}
 & X \arrow[d, dashed, "\Psi"] \arrow[ddl, equal, bend right] \arrow[ddr, bend left] & \\
 & X \overleftarrow{\times}_S S \arrow[dl] \arrow[dr] & \\
X \arrow[dr] & & S \arrow[dl, equal] \\
 & S &
\end{tikzcd}
$$

$\Psi : X \to X \overleftarrow{\times}_S S$ morphism of topos

$\uparrow$

vanishing topos

$\Psi = R\Psi_* : D^+(X, \Lambda) \to D^+(X \overleftarrow{\times}_S S, \Lambda)$

$$
\begin{tikzcd}
 & pt \arrow[d] \arrow[ddl, bend right, "\bar{x}"'] \arrow[ddr, bend left, "\bar{t}"] & \\
 & X \overleftarrow{\times}_S S \arrow[dl] \arrow[dr] & \\
X \arrow[dr, "f"'] \arrow[rr, phantom, "\Leftarrow"] & & S \arrow[dl, equal] \\
 & S &
\end{tikzcd}
$$

pt of $X \overleftarrow{\times}_S S$ $\quad (\bar{x}, \bar{t}, f(\bar{x}) \leftarrow \bar{t})$

$\underline{\text{Th}}$. (Orgogozo) $\exists\, \widetilde{S} \longrightarrow S$ modification s.t.

(1) $\Psi(L|\widetilde{X})$ commutes with base change $T \longrightarrow \widetilde{S}$

(2) $\Psi(L|\widetilde{X}) \in D^b_c(\widetilde{X} \overleftarrow{\times}_{\widetilde{S}} \widetilde{S}, \Lambda)$